

EDUCATION

Rutgers University
Ph.D. in Statistics

Piscataway, NJ
2023–present

**School of the Gifted Young,
University of Science and Technology of China (USTC)**
B.S. in Mathematics, Major GPA: 4.17/4.30

Hefei, China
2019–2023

PREPRINTS/PUBLICATIONS

- Representation Learning

1. Difei Gu, Yunhe Gao, Gerasimos Chatzoudis, **Zihan Dong**, Guoning Zhang, Bangwei Guo, Yang Zhou, Mu Zhou, Dimitris N. Metaxas (2026). LUCID-SAE: Learning Unified Vision-Language Sparse Codes for Interpretable Concept Discovery. Submitted to *ICML 2026*.
2. **Zihan Dong**, Xin Zhou, Ryumei Nakada, Lexin Li, and Linjun Zhang (2025). Contrastive Network Representation Learning. *Submitted to JASA Theory and Methods*. <https://arxiv.org/pdf/2509.11316>.

- Large Language Models

1. **Zihan Dong**, Zhixian Zhang, Yang Zhou, Can Jin, Ruijia Wu, Linjun Zhang (2026). Evaluating LLMs When They Do Not Know the Answer: Statistical Evaluation of Mathematical Reasoning via Comparative Signals. Submitted to *ICML 2026*.
2. Ran Xu, Tianci Liu, **Zihan Dong**, Tony Yu, Ilgee Hong, Carl Yang, Linjun Zhang, Tuo Zhao, Haoyu Wang (2026). Alternating Reinforcement Learning for Rubric-Based Reward Modeling in Non-Verifiable LLM Post-Training. Submitted to *ICML 2026*.
3. Shuhang Lin, Chuha Zhou, Xiao Lin, **Zihan Dong**, Kuan Lu, Zhencan Peng, Jie Yin, Dimitris N. Metaxas (2026). Conformal Path Reasoning: Trustworthy Knowledge Graph Question Answering via Path-Level Calibration. Submitted to *ICML 2026*.
4. Can Jin, Yang Zhou, Qixin Zhang, Hongwu Peng, Di Zhang, **Zihan Dong**, Marco Pavone, Ligong Han, Zhang-Wei Hong, Tong Che, Dimitris N. Metaxas (2026). AIRL-S: Unifying Reinforcement Learning and Search-Based Test-Time Scaling via Adversarial Inverse Reinforcement Learning. Submitted to *ICML 2026*.
5. **Zihan Dong**, Ruijia Wu, and Linjun Zhang (2026). Labels or Preferences? Budget-Constrained Learning with Human Judgments over AI-Generated Outputs. *Submitted to JASA Theory and Methods*. <https://arxiv.org/pdf/2601.13458>.
6. Peng Xia, Jinglu Wang, Yibo Peng, Kaide Zeng, **Zihan Dong**, Xian Wu, Xiangru Tang, Hongtu Zhu, Yun Li, Linjun Zhang, Shujie Liu, Yan Lu, and Huaxiu Yao (2026). MMedAgent-RL: Optimizing Multi-Agent Collaboration for Multimodal Medical Reasoning. *ICLR 2026*.
7. Ran Xu, Yuchen Zhuang, **Zihan Dong**, Ruiyu Wang, Yue Yu, Joyce C. Ho, Linjun Zhang, Haoyu Wang, Wenqi Shi, and Carl Yang (2025). AceSearcher: Bootstrapping Reasoning and Search for LLMs via Reinforced Self-Play. *NeurIPS 2025 Spotlight*.
8. Tianci Liu, Ruirui Li, **Zihan Dong**, Hui Liu, Xianfeng Tang, Qingyu Yin, Linjun Zhang, Haoyu Wang, and Jing Gao (2025). Mitigating Heterogeneous Token Overfitting in LLM Knowledge Editing. *ICML 2025*.

HONORS AND AWARDS

- Best Ph.D. Qualifying Exam Performance Award, Rutgers University

2024

- National Scholarship, Top 1% nationwide, Chinese Ministry of Education 2022
- Cyrus Tang Scholarship, Top 3%, Cyrus Tang Foundation, Nationwide 2019-2022
- The 12th Chinese Mathematics Competitions, Second Prize in Professional Group 2020

TEACHING EXPERIENCE

Instructor

Rutgers University

- Probability and Stochastic Process (for Ph.D student).¹ Summer 2025

Teaching Assistant

Rutgers University

- STATS 654 - Stochastic Process (Core Course for Ph.D. student) Spring 2025
- STATS 382 - Introduction to Theory of Statistics Spring 2025
- STATS 592 - Theory of Probability (Core Course for Ph.D. student) Fall 2024
- STATS 211 - Statistics 1 Spring 2024
- STATS 291 - Statistical Inference for Data Science Spring 2024
- STATS 463 - Regression Methods Fall 2023
- STATS 401 - Basic Statistics for Research Fall 2023

University of Science and Technology of China

- Mathematical Statistics Spring 2022

¹This course is taught by the recipient of the *Best Ph.D. Qualifying Exam Performance Award*.